REPRESENTABILITY BY ALGEBRAIC SPACES IS EQUIVALENT TO THE SECOND DIAGONAL BEING AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The characterisation of morphisms representable by algebraic spaces through faithfulness, and the description of the relative inertia as the kernel of the functor on isomorphism sheaves. Invoked inside the counted proof (source lines 1717-1734) as pointers to results of the chapter on algebraic stacks. Translates representability into a condition on the kernel of the map on Isom sheaves and identifies that kernel with a fibre product of the inertia.
- The description of the second diagonal as the identity section of the relative inertia group stack. The statement uses the second diagonal and the relative inertia interchangeably, and the proof invokes the lemma identifying the kernel of the Isom map with a fibre product of the inertia. Provides the object whose triviality is equivalent to the conditions of the statement.

Source reference. Original reference algebraic-lemma-characterize-representable-by-algebraic-spaces: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1720–1720. Navigation pointer only: the counted text refers to the Stacks item algebraic-lemma-characterize-representable-by-algebraic-spaces, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-inertia: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1327–1327. Navigation pointer only: the counted text refers to the Stacks item lemma-inertia, which belongs to another chapter of the project and is not part of this card. The pointer carries the

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Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent:*

(1) *the morphism f is representable by algebraic spaces,*
(2) *the second diagonal of f is an isomorphism,*
(3) *the group stack $\mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is trivial over $\mathcal{X}$, and*
(4) *for a scheme T and a morphism $x : T \to \mathcal{X}$ the kernel of $Isom_{\mathcal{X}}(x, x) \to Isom_{\mathcal{Y}}(f(x), f(x))$ is trivial.*

Proof. We first prove the equivalence of (1) and (2). Namely, f is representable by algebraic spaces if and only if f is faithful, see Algebraic Stacks, Lemma algebraic-lemma-characterize-representable-by-algebraic-spaces. On the other hand, f is faithful if and only if for every object x of $\mathcal{X}$ over a scheme T the functor f induces an injection $Isom_{\mathcal{X}}(x, x) \to Isom_{\mathcal{Y}}(f(x), f(x))$, which happens if and only if the kernel K is trivial, which happens if and only if $e : T \to K$ is an isomorphism for every $x : T \to \mathcal{X}$. Since $K = T \times_{x,\mathcal{X}} \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ as discussed above, this proves the equivalence of (1) and (2). To prove the equivalence of (2) and (3), by the discussion above, it suffices to note that a group stack is trivial if and only if its identity section is an isomorphism. Finally, the equivalence of (3) and (4) follows from the definitions: in the proof of Lemma lemma-inertia we have seen that the kernel in (4) corresponds to the fibre product $T \times_{x,\mathcal{X}} \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ over T. □